

$$\begin{aligned}
& \text{Maximize} && c_1x_1 + c_2x_2 + c_3x_3 \\
& x \in \mathbb{R}^n && + c_4x_4 + \cdots + c_nx_n \\
& \text{s.t.} && \begin{bmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{m1} & \cdots & a_{mn} \end{bmatrix} \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} \leq \begin{bmatrix} b_1 \\ \vdots \\ b_m \end{bmatrix} \\
& && x \geq 0
\end{aligned}
\tag{1}$$

$$\tag{2}$$